Fiscal Year:	FY 2026	**Task Last Updated:**	FY 02/12/2026
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Project Title:	Composing Teams with TEAMSTaR: Tool for Evaluating and Mitigating Space Team Risk		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-TEAM: Team Composition-Appendix G
Start Date:	04/15/2021	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2025 per NSSC info and NASA-JSC (Ed., 9/10/24) NOTE: End date changed to 4/14/2025 per NSSC info via V. Lehman/JSC Grants Office (Ed., 4/17/24) NOTE: End date changed to 4/14/2024 per NSSC info via L. Barnes-Moten/JSC (Ed., 7/16/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bell, Suzanne Ph.D. (NASA Johnson Space Center) DeChurch, Leslie Ph.D. (Northwestern University, Evanston) Loerch, Linda M.S. (NASA Johnson Space Center)		
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Task Description:	As NASA sets its sight on more Earth-independent missions, such as Artemis missions to the Moon and on to Mars, team composition becomes a critical leverage point for mitigating risks. NASA has successfully designed crews with “the Right Stuff” for more than fifty years beginning with the Mercury and Apollo programs, then into the Shuttle period, and throughout the Skylab and International Space Station missions requiring highly specialized crews to live and work in space for extended periods of time. The key point of departure for deep space exploration is the complexity of missions and the autonomy with which the crew will work. Communication delays with support teams on Earth will necessitate that a relatively small crew take on greater responsibility for making critical decisions. This increased autonomy will occur despite the additional challenges posed by prolonged isolation and confinement and increased radiation exposure. Whereas the “Right Stuff” (Wolfe, 1979) emphasized the requisite individual characteristics, deep space missions also require the “Right Combination” of team members. With that as a backdrop, this project develops and validates TEAMSTaR (Tool for Evaluating And Mitigating Space Team Risks), a team composition decision support system, that can be used by stakeholders (e.g., schedule decision-makers) to predict how a hypothetical team’s social relations are likely to evolve and influence crew performance over the course of a mission. The TEAMSTaR will enable decision-makers to evaluate composition scenarios for an entire set of teams, for single-member replacements, and/or for subsets of teams. To do this, we first leverage insights and data from recent NASA-funded team composition studies and thoughtfully refine and extend our agent-based models to include relevant input characteristics and their ability to predict team outcomes including team performance. We next conduct virtual experiments and gather stakeholder input to inform the development of TEAMSTaR, a team composition decision support system that utilizes insights from our updated agent-based models (ABMs) to enable real-time (or close to real-time) decision-making. Finally, we validate TEAMSTaR as a decision-making tool in short and long-term isolated, confined, and controlled environments. This project accomplishes five aims. Aim 1) Refine agent-based models looking at relevant input characteristics and their ability to predict team outcomes, including team performance. Aim 2) Identify and elaborate the scientific rationale for attributes used within the model, identifying factors known to affect crew functioning, crew member behavior, emergent characteristics that arise during team task completion. Aim 3) Develop and validate a Team Composition decision support system and user interface. Aim 4) Validate the refined model using a software prototype in at least one extended duration, isolated, and confined analog. Aim 5) Provide modeling and software prototypes that meet NASA Standard 7009a. With an eye toward the future of deep space exploration, this project leverages, and indeed advances, state of the art computational techniques to predict crew performance and to identify points of leverage in terms of team composition and task scheduling to optimize individual and team performance. Reference: Wolfe, T. (1979). The Right Stuff. New York: Farrar, Straus and Giroux.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With an eye toward the future of deep space exploration, this project leverages and advances state of the art computational techniques to predict crew performance and to identify points of leverage in terms of team composition and task scheduling to optimize individual and team performance. Even though TEAMSTaR decision support system will be tested in space analog missions, it can be applied to teams operating on Earth in isolated and confined environments (ICE), such as expedition and science teams in the Arctic and Antarctic. The general framework of team composition and the analytic strategies developed in this project can be applied to Earth teams more generally.
Task Progress:	Over the course of this project, our team successfully developed, refined, and validated TEAMSTaR, a new decision support system designed to help NASA crews maintain strong teamwork and performance during long-duration space missions. The work brought together advances in team science, computational modeling, and human-in-the-loop testing to create a functional prototype that can support both astronauts and mission support personnel. A major accomplishment of the project was the creation of the TEAMSTaR dashboard, an interactive tool that allows crews and Ground Support Personnel to view predictions about team functioning and explore how different decisions may influence teamwork and task performance. The dashboard was designed through an iterative, user-centered process, beginning with early mock-ups and culminating in a fully operational prototype. Alongside the interface, we developed a comprehensive set of surveys, team-building activities, and operational tasks that allowed us to test the system in a realistic mission environment. Another key achievement was the development of an integrated computational model that predicts how team relationships and performance may evolve over time. This model draws on more than a decade of NASA-supported research and incorporates factors such as leadership patterns, communication dynamics, shared mental models, and interpersonal interactions. The model was used to generate weekly simulations of crew networks and performance projections, which were then presented to crews during mission activities. The TEAMSTaR system was fully implemented and tested in NASA’s Human Exploration Research Analog (HERA) Campaign 7. Across four missions, crews used the dashboard each week to review predictions, make decisions about team-building activities, and complete operational tasks in pairs. This provided a rich dataset on how crews interacted with the system, how they responded to different types of decisions, and how TEAMSTaR’s predictions aligned with real-world outcomes. We collected data on user experience, teamwork, and task performance, allowing us to evaluate the system’s usefulness and identify areas for improvement. The project also produced several important findings about how to support crew functioning in isolated and confined environments. We found that simple, low-burden team-building activities were well-received and tended to improve teamwork. Crews reported higher satisfaction when they were able to choose their own team-building activities, suggesting that autonomy plays a meaningful role in engagement. In contrast, operational task assignments were more effective when selected by Ground Support Personnel, who may have a broader perspective on workload and mission needs. These insights offer practical guidance for balancing crew autonomy with ground-directed structure in future missions. In addition to developing and testing the system, we completed all required analyses and prepared the results for

	dissemination. We examined how TEAMSTaR influenced satisfaction, teamwork, and performance, and we identified opportunities to refine the dashboard to improve usability and clarity. The project also contributed to scientific outreach through conference presentations, workshops, and peer-reviewed publications, helping to share lessons learned with NASA stakeholders and the broader research community. By the end of the funding period, all project aims were completed. TEAMSTaR now exists as a functional prototype that integrates advanced modeling, user-friendly design, and real-world testing. The system provides a foundation for future development and represents a promising step toward supporting astronaut autonomy and team resilience during long-duration exploration missions.
Bibliography Type:	Description: (Last Updated: 03/17/2026)
Articles in Peer-reviewed Journals	Lungeanu A, DeChurch L, Chan M, Contractor N. "Leading the crew to Mars: Evidence from NASA HERA analog crews." Acta Astronaut. 2025 Aug 16. https://doi.org/10.1016/j.actaastro.2025.08.030 , Aug-2025
Articles in Peer-reviewed Journals	Carter D, Lungeanu A, Pearman J, Carter N, Bauer L, Pendergraft J, Landon LB, Shuffler M, DeChurch L, Contractor N. "Collective attention in virtual teams: A pathway for mitigating communication delays." Pers. Psychol. 2026 Mar 2. Online ahead of print. https://doi.org/10.1111/peps.70024 , Mar-2026